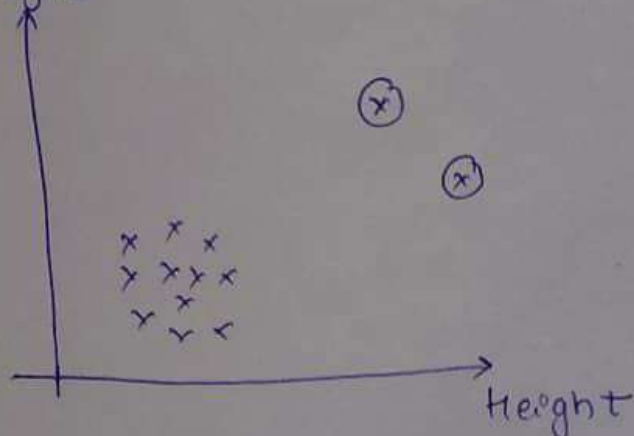


Anomaly Detection [To detect Outliers]

weight

→ play a important Role



1PL

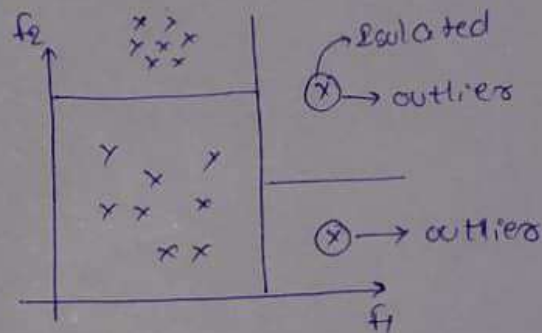
1	15
2	10
3	12

4 **100** **> 36**

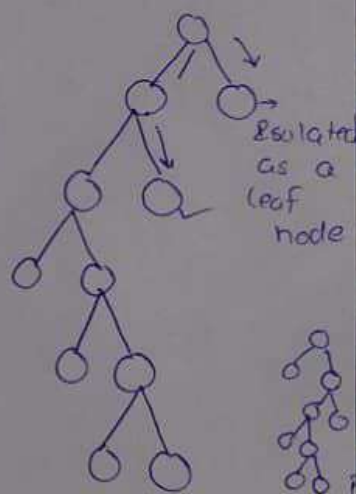
Many Trees

① Isolation Forest [Decision Trees]

f_1	f_2	f_3	f_4
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-



Isolated Tree



Anomaly score

Mathematical formula :- Compute

anomaly score for a new point

$$S(\alpha, m) = 2^{-\frac{E(h(\alpha))}{c(m)}}$$

m = no. of data points

α = Data point

$E(h(\alpha))$ = Average search depth for α from the isolated tree

$c(m)$ = Average value of $h(\alpha)$ [Threshold ≥ 0.5]

$E(h(\alpha)) \ll c(m) \Rightarrow S(\alpha, m) \approx 1 \Rightarrow$ Average score $\Rightarrow$ outliers

$E(h(\alpha)) \gg c(m) \Rightarrow S(\alpha, m) \approx 0.5 \Rightarrow$ Normal data point